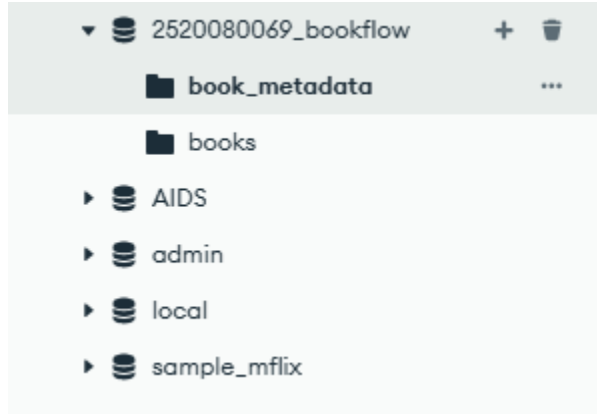


PRACTICAL WEEK3 : MongoDB Atlas

PART 1 : Basics

Query 1 : Create Database and collections

OUTPUT



Query 2 : Insert documents

```
{
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101,
  "reviews": [
    {
      "member_id": 101,
      "rating": 5,
      "comments": "Excellent science fiction story"
    },
    {
      "member_id": 102,
      "rating": 4,
      "comments": "Very interesting and informative"
    },
    {
      "member_id": 103,
      "rating": 5,
      "comments": "Excellent space adventure"
    }
  ]
}
```

}
]
}
OUTPUT:

ORGANIZATION
Ananya Maroj's Org...

PROJECT
2520080069_Ahanya

My Queries

Data Modelling

CLUSTERS (0)

Search clusters

Cluster0

- 2520080069_bookflow
 - book_metadata
 - books
 - AIDS
 - admin
 - local
 - sample_mflix

Cluster0 > 2520080069_bookflow > book_metadata

Documents

Aggregations

Schema

Indexes

Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

251 - 5 of 5

	book_metadata	_id ObjectId	book_id Int32	title String	author String	published_year Int32	category String	format String	
1	ObjectId('6a8d1ea398b81f...	101		"The Martian"	"Andy Weir"	2021	"Science Fiction"	"Digital PDF"	/🔍🔍🔍
2	ObjectId('6a8d1f1098b81f...	102		"Dune"	"Frank Herbert"	2022	"Science Fiction"	"Digital EPUB"	/🔍🔍🔍
3	ObjectId('6a8d1f1c98b81f...	103		"Clean Code"	"Robert C. Martin"	2019	"Programming"	"Physical"	/🔍🔍🔍
4	ObjectId('6a8d1f2f98b81f...	104		"Artificial Intelligence...	"John Smith"	2023	"Artificial Intelligence"	"Digital PDF"	/🔍🔍🔍
5	ObjectId('6a8d1f3c98b81f...	105		"Data Structures"	"Mark Allen"	2024	"Computer Science"	"Digital PDF"	/🔍🔍🔍

Alert System Status: All Good

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Query 3 : Comparison Operator

```
{
  "reviews.rating": {
    "$gt": 4
  }
}
```

OUTPUT :

ORGANIZATION
Ananya Maraj's Org...

PROJECT
2520080069_Ananya

My Queries

Data Modelling

CLUSTERS (1)

Cluster0

book_metadata

books

AIDS

admin

local

sample_mflix

Customerbooksbook_metadata

Cluster0 > 2520080069_bookflow > book_metadata

Documents5AggregationsSchemaIndexes1Validation

Generate query

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	category String	format String	
1	ObjectId("6a8d1ea398b81f...	181	"The Martian"	"Andy Weir"	2021	"Science Fiction"	"Digital PDF"	
2	ObjectId("6a8d1f098b81f...	182	"Dune"	"Frank Herbert"	2022	"Science Fiction"	"Digital EPUB"	
3	ObjectId("6a8d1f1c98b81f...	183	"Clean Code"	"Robert C. Martin"	2019	"Programming"	"Physical"	
4	ObjectId("6a8d1f2f98b81f...	184	"Artificial Intelligence...	"John Smith"	2023	"Artificial Intelligence"	"Digital PDF"	
5	ObjectId("6a8d1f3c98b81f...	185	"Data Structures"	"Mark Allen"	2024	"Computer Science"	"Digital PDF"	

Query 4 : Using \$IN

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

OUTPUT:

The screenshot shows the Data Explorer interface for a project named '2520080069_Ananya'. The left sidebar displays the 'Data Explorer' view with a tree structure showing clusters and data sources. The main area shows a query for 'Cluster0' with the following JSON:

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

The query results are displayed in a table with 7 columns: `_id`, `book_id`, `title`, `author`, `published_year`, `category`, and `format`. The table contains 5 rows of data.

#	book_metadata	_id	book_id	title	author	published_year	category	format
1	ObjectId('6a8d1ea398b81f...	101	"The Martian"	"Andy Weir"	2021	"Science Fiction"	"Digital PDF"	📄 🔄 🗑️
2	ObjectId('6a8d1f1898b81f...	102	"Dune"	"Frank Herbert"	2022	"Science Fiction"	"Digital EPUB"	📄 🔄 🗑️
3	ObjectId('6a8d1f1c98b81f...	103	"Clean Code"	"Robert C. Martin"	2019	"Programming"	"Physical"	📄 🔄 🗑️
4	ObjectId('6a8d1f2f98b81f...	104	"Artificial Intelligence...	"John Smith"	2023	"Artificial Intelligence"	"Digital PDF"	📄 🔄 🗑️
5	ObjectId('6a8d1f3c98b81f...	105	"Data Structures"	"Mark Allen"	2024	"Computer Science"	"Digital PDF"	📄 🔄 🗑️

Query 5 : Regex query

```
{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

OUTPUT:

The screenshot shows the Data Explorer interface with the following components:

- Left Panel:** Contains 'Data Explorer', 'My Queries', 'Data Modeling', and 'CLUSTERS (1)'. The 'CLUSTERS (1)' section shows a search bar and a list of clusters including 'Cluster0' and '2520080049_bookflow'.
- Top Panel:** Displays the breadcrumb 'Cluster0 > 2520080049_bookflow > book_metadata' and buttons for 'View monitoring' and 'Visualize Your Data'.
- Query Editor:** Shows a JSON query:

```
{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

 with buttons for 'Generate query', 'Explain', 'Reset', 'Find', and 'Options'.
- Results Table:** Displays a table with columns: '_id ObjectId', 'book_id Int32', 'title String', 'author String', 'published_year Int32', 'category String', and 'format String'. It contains 4 rows of data.

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	category String	format String
1	ObjectId('6a8d1ea398b81f...')	181	"The Martian"	"Andy Weir"	2021	"Science Fiction"	"Digital PDF"
2	ObjectId('6a8d1f198b81f...')	182	"Dune"	"Frank Herbert"	2022	"Science Fiction"	"Digital EPUB"
3	ObjectId('6a8d1f2f98b81f...')	184	"Artificial Intelligence..."	"John Smith"	2023	"Artificial Intelligence"	"Digital PDF"
4	ObjectId('6a8d1f3c98b81f...')	185	"Data Structures"	"Mark Allen"	2024	"Computer Science"	"Digital PDF"

PART 2 : Aggregate pipeline

Query 6 : \$MATCH

Stage:

```
{
  "$match": {
    "published_year": {
      "$gt": 2020
    }
  }
}
```

OUTPUT:

Cluster02520080069_bookflowbook_metadata

Documents5

Aggregations

Schema

Indexes1

Validation

View monitoring

Visualize Your Data

Documents

\$match

Generate aggregation

Explain

Run

Options

Untitled - modified

SAVE

CREATE NEW

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Preview of documents

Stage1\$match

1/**

2* query: The query in MQL.

3*/

4{

5 "published_year": {

6 "\$gt": 2020

7 }

8}

9

Output preview after \$match stage (Sample of 4 documents)

_id: ObjectId('6a8d1ea398b81f4a9db9cd8f')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

_id: ObjectId('6a8d1f098b81f4a9db9cd91')

book_id: 102

title: "Dune"

author: "Frank Herbert"

published_year: 2022

category: "Science Fiction"

format: "Digital EPUB"

file_size: "3.1 MB"

member_id: 102

_id: ObjectId('6a8d1f2f98b81f4a9db9cd95')

book_id: 104

title: "Artificial Intelligence Basics"

author: "John Smith"

published_year: 2023

category: "Artificial Intelligence"

format: "Digital PDF"

file_size: "4.2 MB"

member_id: 104

+ Add stage

Learn more about aggregation pipeline stages

System Status: All Good

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Query 7 : \$UNWIND

```
{
  "$unwind": "$reviews"
}
```

OUTPUT:

Cluster02520080069_bookflowbook_metadata

Documents5

Aggregations

Schema

Indexes1

Validation

View monitoring

Visualize Your Data

Documents

\$match

\$unwind

\$group

\$sort

Generate aggregation

Explain

Run

Options

Untitled - modified

SAVE

CREATE NEW

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Preview of documents

Stage2\$unwind

1

2 "\$reviews"

3

Output preview after \$unwind stage (Sample of 10 documents)

_id: ObjectId('6a8d1ea398b81f4a9db9cd8f')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

_id: ObjectId('6a8d1ea398b81f4a9db9cd8f')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

_id: ObjectId('6a8d1ea398b81f4a9db9cd8f')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

Query 8 : \$GROUP

```
{
  "$group": {
    "_id": "$title",
    "average_rating": {
      "$avg": "$reviews.rating"
    },
    "total_reviews": {
      "$sum": 1
    }
  }
}
```

OUTPUT:

Cluster0 > 2620080049_bookflow > book_metadata

Documents (5) Aggregations Schema Indexes (1) Validation

Match Sunwind \$group \$sort Generate aggregation + Explain Run Options

Untitled - modified SAVE + CREATE NEW EXPORT CODE PREVIEW STAGES TEXT WIZARD

Stage 3 \$group

```
1 //
2 * _id: The id of the group.
3 * fieldN: The first field name.
4 */
5 {
6   "_id": "$title",
7   "average_rating": {
8     "$avg": "$reviews.rating"
9   },
10  "total_reviews": {
11    "$sum": 1
12  }
13 }
```

Output preview after \$group @ stage (Sample of 4 documents)

<pre>{ "_id": "Dune" "average_rating": 4.666666666666667 "total_reviews": 3 }</pre>	<pre>{ "_id": "Artificial Intelligence Basics" "average_rating": 4.333333333333333 "total_reviews": 3 }</pre>	<pre>{ "_id": "The Martian" "average_rating": 4.666666666666667 "total_reviews": 3 }</pre>
---	---	--

Query 9 : \$SORT

```
{
  "$sort": {
    "average_rating": -1
  }
}
```

OUTPUT:

The screenshot shows the MongoDB Compass interface for a cluster named 'Cluster0' and a database named '2520080069_bookflow'. The 'book_metadata' collection is selected. The 'Aggregations' tab is active, showing a single stage named '\$sort'. The stage configuration is set to 'Sort' with a sort order of '-1' for the 'average_rating' field. The output preview shows three documents sorted by 'average_rating' in descending order:

```
{ "_id": "The Martian", "average_rating": 4.666666666666667, "total_reviews": 3 }
{ "_id": "Dune", "average_rating": 4.333333333333333, "total_reviews": 3 }
{ "_id": "Artificial Intelligence Basics", "average_rating": 4.333333333333333, "total_reviews": 3 }
```

PART 3 : Text Index

Query 10 : Creation of text index

```
db.book_metadata.createIndex(
{
  "reviews.comments": "text"
})
```

OUTPUT:

The screenshot shows the MongoDB Compass interface for the same cluster and database. The 'Indexes' tab is active, showing a list of indexes for the 'book_metadata' collection. A new index has been created for the 'reviews.comments' field with a 'TEXT' type. The index is named '_id_'. The status of the index is 'READY'.

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	36.9 KB	7 (since Tue Aug 25 2026)	UNIQUE	READY
> reviews.comments_text	TEXT	20.5 KB	0 (since Tue Aug 25 2026)		READY

Query 11 : Search using text index

```
{
  "$text": {
    "$search": "excellent"
  }
}
```

OUTPUT:

Cluster0 > 2520080049_bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	category String	format String	file_s
1	ObjectId('6a8d1ea398b81f...	101	"The Martian"	"Andy Weir"	2021	"Science Fiction"	"Digital PDF"	"2.5 / / / / /"
2	ObjectId('6a8d1f1c98b81f...	103	"Clean Code"	"Robert C. Martin"	2019	"Programming"	"Physical"	No fic / / / / /"
3	ObjectId('6a8d1f1098b81f...	102	"Dune"	"Frank Herbert"	2022	"Science Fiction"	"Digital EPUB"	"3.1 / / / / /"
4	ObjectId('6a8d1f3c98b81f...	105	"Data Structures"	"Mark Allen"	2024	"Computer Science"	"Digital PDF"	"5.0 / / / / /"
5	ObjectId('6a8d1f2f98b81f...	104	"Artificial Intelligence..."	"John Smith"	2023	"Artificial Intelligence"	"Digital PDF"	"4.2 / / / / /"

Query 12 : Search for a specific keyword

```
{
  "$text": {
    "$search": "\"science fiction\""
  }
}
```

OUTPUT:

Cluster0 > 2520080049_bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 2 of 2

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	category String	format String	file_s
1	ObjectId('6a8d1f1098b81f...	102	"Dune"	"Frank Herbert"	2022	"Science Fiction"	"Digital EPUB"	"3.1 / / / / /"
2	ObjectId('6a8d1ea398b81f...	101	"The Martian"	"Andy Weir"	2021	"Science Fiction"	"Digital PDF"	"2.5 / / / / /"

Query 13 : USE explain()

Check whether MongoDB is using an index or performing a collection scan.

```
{
  "$text": {
    "$search": "excellent"
  }
}
```

OUTPUT:

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

[Visual tree](#) [Raw Output](#)

```
graph TD
    A[TEXT_MATCH] --> B[FETCH]
    B --> C[IXSCAN]
```

The diagram illustrates the execution plan for the query. It consists of three stages: TEXT_MATCH, FETCH, and IXSCAN. Each stage shows the number of documents returned and the execution time. The IXSCAN stage indicates that the index used is reviews.comments... and it is a multi-key index.

Query Performance Summary

- 6 documents returned
- 6 documents examined
- 0 ms execution time
- is not sorted in memory
- 6 index keys examined

Query used the following index:

- fts (text)
- ftsx